Practice workbook · Explain causes of obesity, its health risks and dietary advice

Energy imbalance causing obesity, its links to diabetes and heart disease, and diet changes to reduce mass.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-OBESITY-CAUSES-RISKS

1. Question 4 · part a.i · 3 marks

June 2021 Paper 2

Worked-example retry · Attempt part a.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

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(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

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1. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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2. Question 5 · part b.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.ii only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

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mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

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2 .....

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(iii) Give two ways the teacher could improve his investigation.

(2)

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2. Question 5 · part b.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.ii only.

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(b) The table shows data for two different students.

Student	Height in m	Mass in kg	BMI
X	1.69	62	21.7
Y	1.46	71	

(i) BMI is calculated using the equation

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

Calculate the BMI of student Y.
[mass measured in kg, height measured in m]

(2)

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BMI of student Y = ...

(ii) BMI can be used to assess whether a person has a healthy mass for their height.

BMI	Classification
below 18.5	underweight
18.5 – 24.9	normal weight
25 – 29.9	overweight
30 – 40	obese
above 40	morbidly obese

Using information from the table, explain how the BMI of student Y could affect her health.

(2)

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(Total for Question 5 = 10 marks)

2. Question 5 · part b.ii · 2 marks · Working and answer

Working and answer

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3. Question 1 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

1 (a) Food packaging gives information about the food it contains.

The photograph shows some of the information given on a package containing yogurt.



(Source: © studiomode / Alamy Stock Photo)

The instructions state that the yogurt should be stored in a fridge at 4°C.

Why should this instruction be followed?

(1)

- ☒ **A** to make the yogurt taste better
- ☒ **B** to kill all bacteria in the yogurt
- ☒ **C** to slow down the growth of bacteria in the yogurt
- ☒ **D** to improve the appearance of the yogurt

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3. Question 1 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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(b) The diagram shows the nutritional label on a package containing yogurt.

Nutrition Facts
Serving Size
1 container (150g)
Energy in kJ 480
Energy from Fat 0

Amount/Serving	% Daily Value*	Amount/Serving	% Daily Value*
Total Fat	a%	Sodium	2%
Saturated Fat	0%	Total Carb.	6%
Trans Fat		Sugars	
Cholesterol	1%	Protein	22%

Vitamin A 4% • Calcium 10% • Vitamin D 20%

Not a significant source of dietary fibre, vitamin C and iron.

(i) State which nutrient on the label is required for growth and repair of body tissue.

(1)

.....

(ii) Why is this yogurt suitable for a person who wants to reduce their body mass?

(1)

- ☒ **A** it contains vitamins C and D
- ☒ **B** it contains less than 5 mg of cholesterol
- ☒ **C** it does not contain energy from carbohydrates
- ☒ **D** it does not contain energy from fat

(c) Yogurt contains reducing sugars.

Describe a method to test for reducing sugars.

(3)

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(Total for Question 1 = 6 marks)

3. Question 1 · part b.ii · 1 mark · Working and answer

Working and answer

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4. Question 4 · part a.ii · 2 marks

June 2021 Paper 2

Further practice · Attempt part a.ii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

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(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

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4. Question 4 · part a.ii · 2 marks

June 2021 Paper 2

Further practice · Attempt part a.ii only.

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ...%

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

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4. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 3 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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5. Question 3 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

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(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

5. Question 3 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

(f) The passage states that how much we sweat varies from individual to individual and from day to day.

Explain why this is the case in people without hyperhidrosis (lines 8 and 9).

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(Total for Question 3 = 14 marks)

5. Question 3 · part e · 1 mark · Working and answer

Working and answer

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6. Question 3 · part c.ii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

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6. Question 3 · part c.ii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

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(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

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(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

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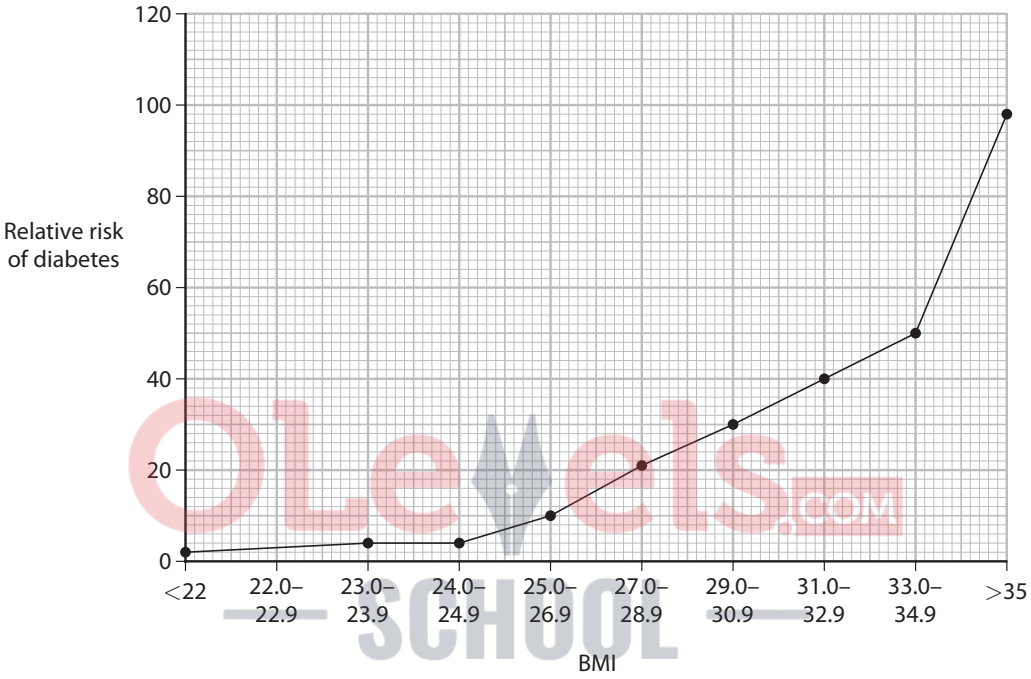
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6. Question 3 · part c.ii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

(c) The graph shows the relative risk of developing a form of diabetes called type 2 for different BMI values.



(Source: adapted from Colditz GA, et al., *Ann Intern Med.*, 1995; 122:481–486)

(i) Describe the trend shown by the graph.

(2)

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6. Question 3 · part c.ii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

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(ii) Give some dietary advice that could help an obese person lower their BMI.

(2)

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(Total for Question 3 = 11 marks)

Working and answer

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